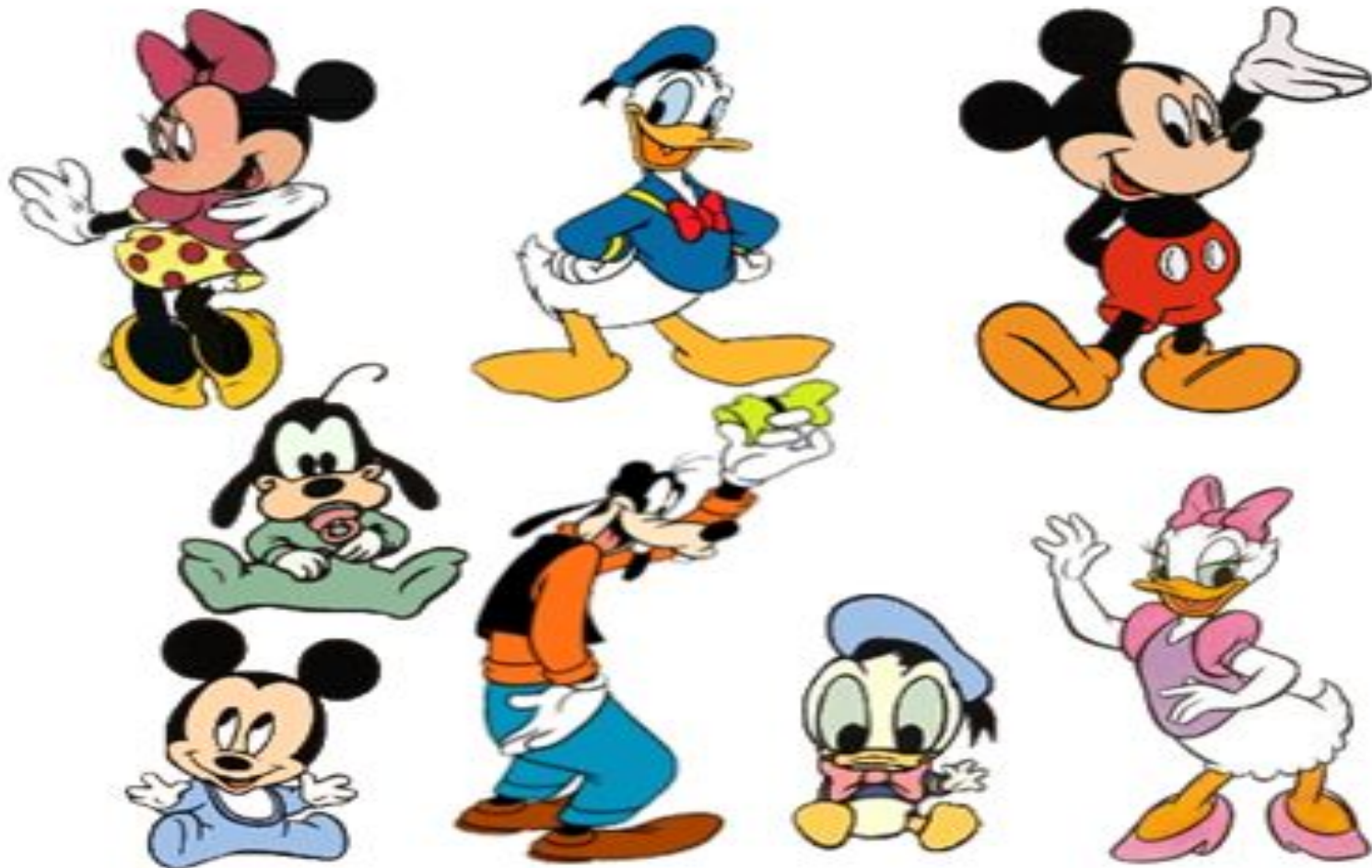


What is Cluster Analysis?

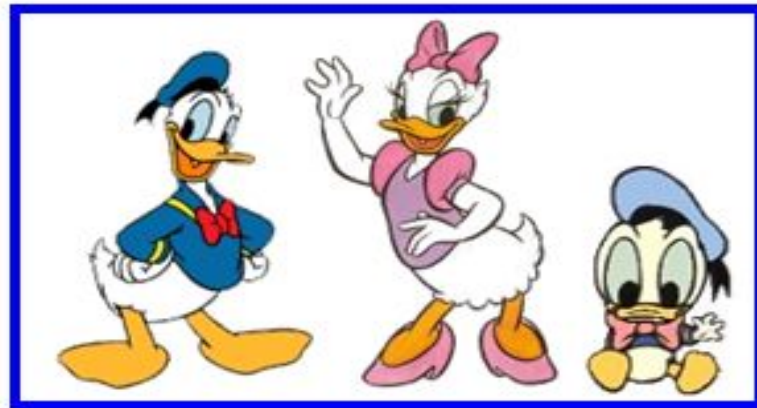
- ❖ **Cluster**: A collection of data objects
 - **similar** (or related) to one another within the **same group**
 - **dissimilar** (or unrelated) to the objects in **other groups**
- ❖ **Cluster analysis** (or *clustering*, *data segmentation*, ...)
 - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters
- ❖ **Unsupervised learning**: no predefined classes (i.e., *learning by observations* vs. learning by examples: supervised)
- ❖ Typical applications
 - As a **stand-alone tool** to get insight into data distribution
 - As a **preprocessing step** for other algorithms

What is Cluster Analysis?



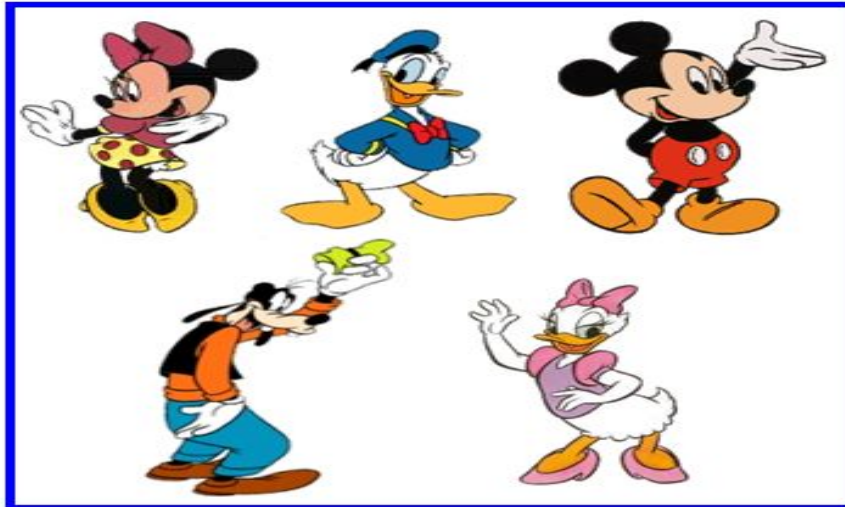
What is Cluster Analysis?

❖ By Family



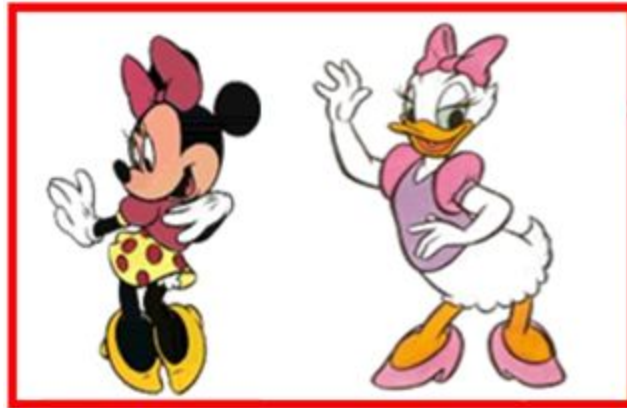
What is Cluster Analysis?

❖ By Age



What is Cluster Analysis?

❖ By Gender



Applications of Cluster Analysis

- ❖ Data **reduction**
 - **Summarization**: Preprocessing for regression, PCA, classification, and association analysis
 - **Compression**: Image processing: vector quantization
- ❖ Hypothesis generation and testing
- ❖ Prediction based on groups
 - Cluster & find characteristics/patterns for each group
- ❖ Finding K-nearest Neighbors
 - Localizing search to one or a small number of clusters
- ❖ **Outlier detection**: Outliers are often viewed as those “far away” from any cluster

Clustering: Application Examples

- ❖ **Biology**: taxonomy of living things: kingdom, phylum, class, order, family, genus and species
- ❖ **Information retrieval**: document clustering
- ❖ **Land use**: Identification of areas of similar land use in an earth observation database
- ❖ **Marketing**: Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- ❖ **City-planning**: Identifying groups of houses according to their house type, value, and geographical location
- ❖ **Earth-quake studies**: Observed earth quake epicenters should be clustered along continent faults
- ❖ **Climate**: understanding earth climate, find patterns of atmospheric and ocean
- ❖ **Economic Science**: market research

Earthquakes

GLOBAL SEISMIC HAZARD MAP

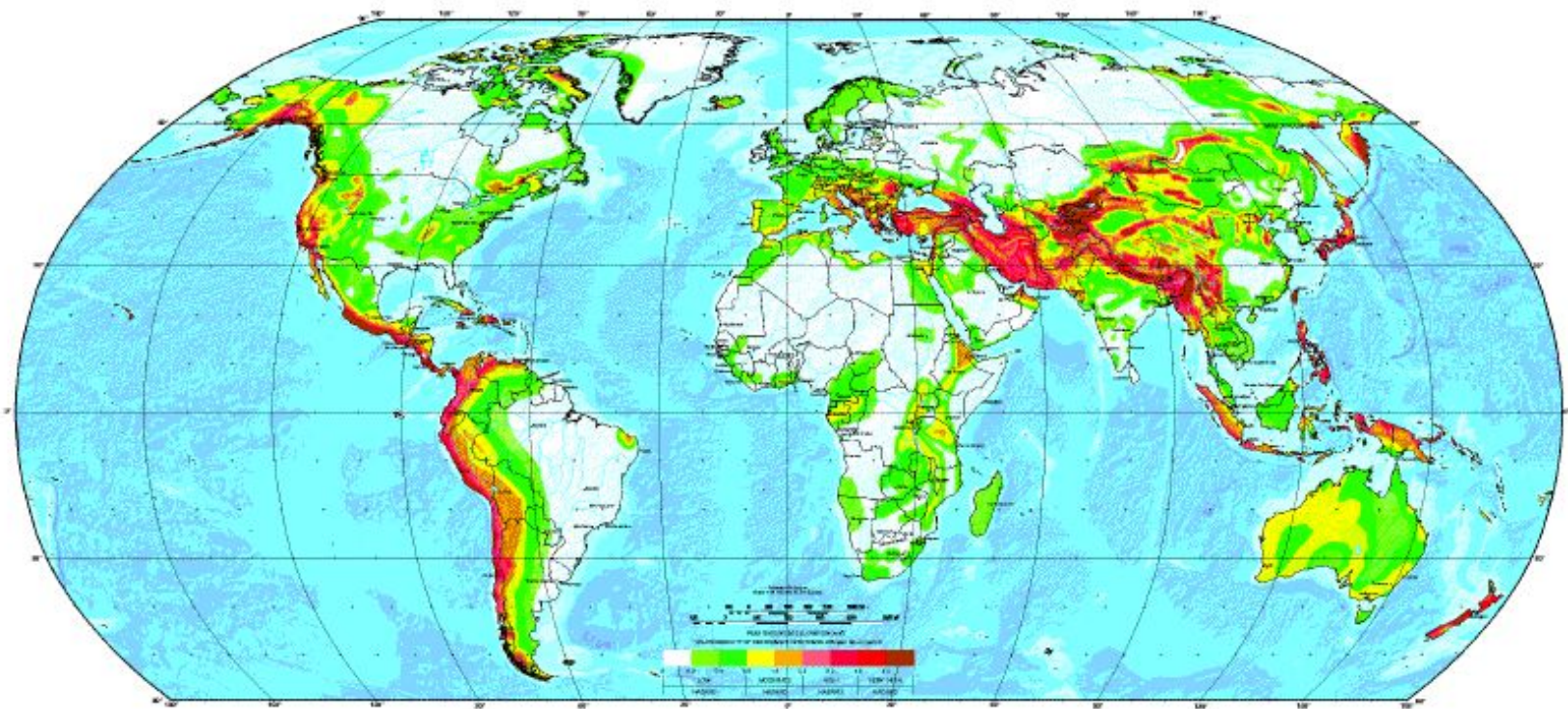


Image Segmentation

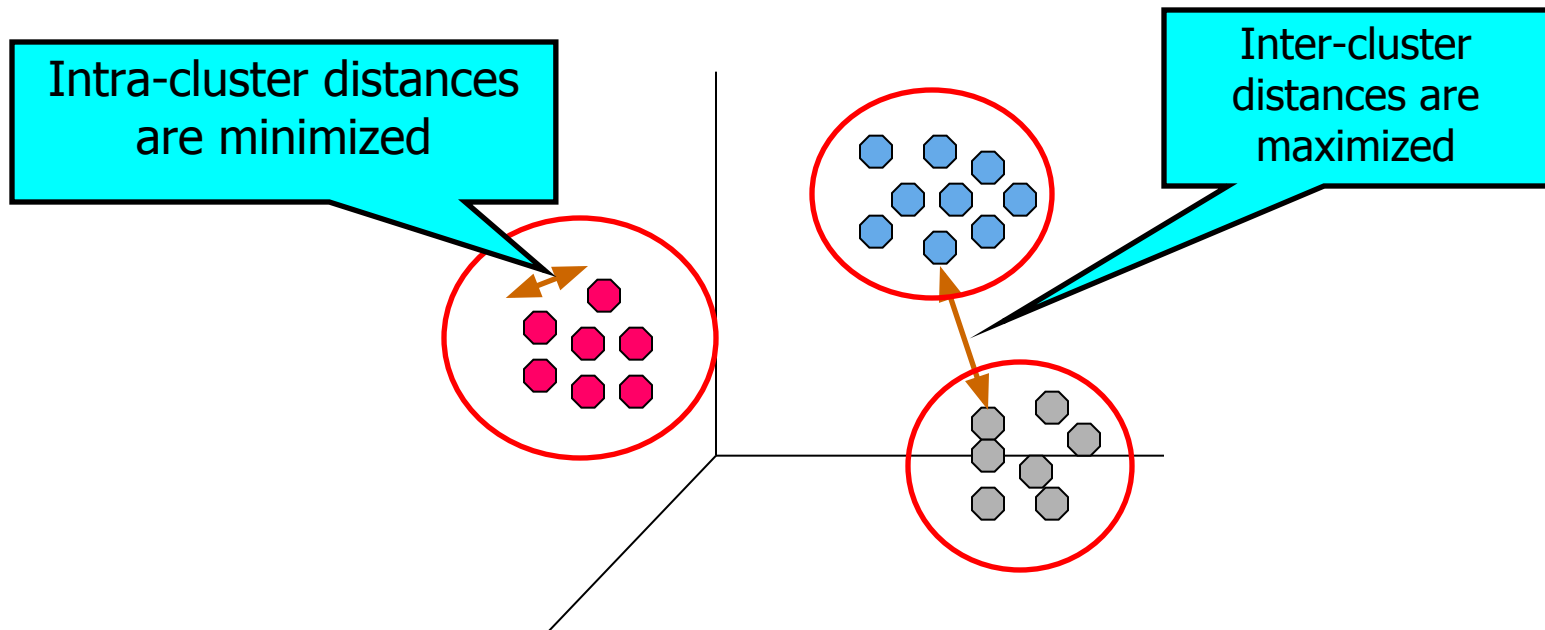


Basic Steps to Develop a Clustering Task

- ❖ Feature selection
 - Select info concerning the task of interest
 - Minimal information redundancy
- ❖ Proximity measure
 - Similarity of two feature vectors
- ❖ Clustering criterion
 - Expressed via a cost function or some rules
- ❖ Clustering algorithms
 - Choice of algorithms
- ❖ Validation of the results
 - Validation test (also, *clustering tendency* test)
- ❖ Interpretation of the results
 - Integration with applications

Quality: What Is Good Clustering?

- ❖ A good clustering method will produce high quality clusters
 - high intra-class similarity: cohesive within clusters
 - low inter-class similarity: distinctive between clusters
- ❖ The quality of a clustering method depends on
 - the similarity measure used by the method
 - its implementation, and
 - Its ability to discover some or all of the hidden patterns



Measure the Quality of Clustering

- ❖ **Dissimilarity/Similarity metric**
 - Similarity is expressed in terms of a distance function, typically metric: $d(i, j)$
 - The definitions of **distance functions** are usually rather different for interval-scaled, boolean, categorical, ordinal ratio, and vector variables
 - Weights should be associated with different variables based on applications and data semantics
- ❖ **Quality of clustering:**
 - There is usually a separate “quality” function that measures the “goodness” of a cluster.
 - It is hard to define “similar enough” or “good enough”
 - The answer is typically highly subjective

Considerations for Cluster Analysis

❖ Partitioning criteria

- Single level vs. hierarchical partitioning (often, multi-level hierarchical partitioning is desirable)

❖ Separation of clusters

- Exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class)

❖ Similarity measure

- Distance-based (e.g., Euclidian, road network, vector) vs. connectivity-based (e.g., density or contiguity)

❖ Clustering space

- Full space (often when low dimensional) vs. subspaces (often in high-dimensional clustering)

Requirements and Challenges

- ❖ Scalability
 - Clustering all the data instead of only on samples
- ❖ Ability to deal with different types of attributes
 - Numerical, binary, categorical, ordinal, linked, and mixture of these
- ❖ Constraint-based clustering
 - User may give inputs on constraints
 - Use domain knowledge to determine input parameters
- ❖ Interpretability and usability
- ❖ Others
 - Discovery of clusters with arbitrary shape
 - Ability to deal with noisy data
 - Incremental clustering and insensitivity to input order
 - High dimensionality

Major Clustering Approaches 1

- ❖ **Partitioning** approach:
 - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: **k-means**, k-medoids, CLARANS
- ❖ **Hierarchical** approach:
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: Diana, Agnes, BIRCH, CHAMELEON
- ❖ **Density-based** approach:
 - Based on connectivity and density functions
 - Typical methods: DBSACN, OPTICS, DenClue
- ❖ **Grid-based** approach:
 - based on a multiple-level granularity structure
 - Typical methods: STING, WaveCluster, CLIQUE

Major Clustering Approaches 2

- ❖ **Model-based:**
 - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
 - Typical methods: EM, SOM, COBWEB
- ❖ **Frequent** pattern-based:
 - Based on the analysis of frequent patterns
 - Typical methods: p-Cluster
- ❖ **User-guided** or constraint-based:
 - Clustering by considering user-specified or application-specific constraints
 - Typical methods: COD (obstacles), constrained clustering
- ❖ **Link-based** clustering:
 - Objects are often linked together in various ways
 - Massive links can be used to cluster objects: SimRank, LinkClus

Partitioning Algorithms: Basic Concept

- ❖ **Partitioning method**: Partitioning a database D of n objects into a set of k clusters, such that the sum of squared distances is minimized (where c_i is the centroid or medoid of cluster C_i)

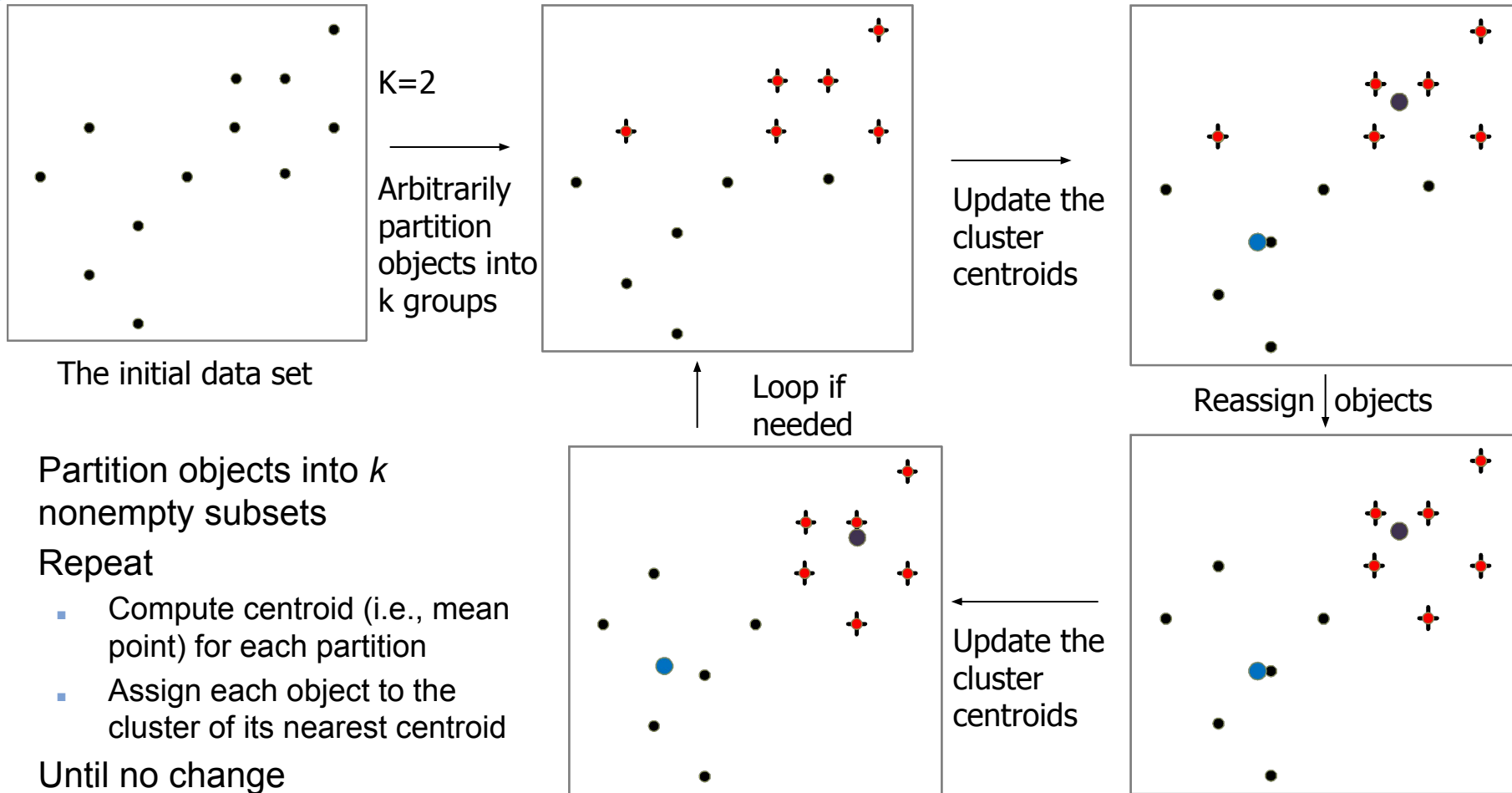
$$E = \sum_{i=1}^k \sum_{p \in C_i} (d(p, c_i))^2$$

- ❖ Given k , find a partition of k clusters that optimizes the chosen partitioning criterion
 - **Global optimal**: exhaustively enumerate all partitions
 - **Heuristic methods**: *k-means* and *k-medoids* algorithms
 - *k-means* (MacQueen'67, Lloyd'57/'82): Each cluster is represented by the center of the cluster
 - *k-medoids* or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

The *K-Means* Clustering Method

- ❖ Given k , the *k-means* algorithm is implemented in four steps:
 1. Partition objects into k nonempty subsets
 2. Compute seed points as the centroids of the clusters of the current partitioning (the centroid is the center, i.e., *mean point*, of the cluster)
 3. Assign each object to the cluster with the nearest seed point
 4. Go back to Step 2, stop when the assignment does not change

An Example of K-Means Clustering



k-Means Algorithms steps

1. Selecting the desired **number of clusters k**
2. **Initialization k cluster centers** (centroid) random
3. Place any data or object to the nearest cluster. The proximity of the two objects is determined based on the distance. The distance used in k-means algorithm is the Euclidean distance (d)

$$d_{Euclidean}(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$x = x_1, x_2, \dots, x_n$ and $y = y_1, y_2, \dots, y_n$ n is the number of attributes (columns) between 2 records

4. **Recalculate cluster centers** with the current cluster membership. Cluster center is the average (mean) of all the data, or objects in a particular cluster
5. **Assign each object again using the new cluster centers.** If the cluster centers have not changed again, then the clustering process is completed. Or, go back to step 3 until the cluster centers do not change anymore (stable) or no significant decrease of the value of SSE (Sum of Squared Errors)

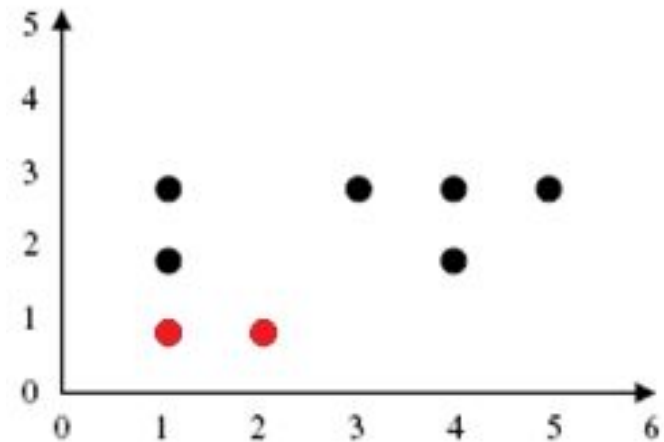
Exercise – Iteration 1

[STEP 1] Determine the number of clusters $k = 2$

[STEP 2] Determine the initial centroid random example of the data
besides $m1 = (1,1)$, $m2 = (2,1)$

[STEP 3] Place each object to the closest cluster centroid based on the
closest value difference (distance).

Instances	x	y	m1	m2	Cluster Membership
			(1,1)	(2,1)	
A	1	3	2	3	m1
B	3	3	4	3	m2
C	4	3	5	4	m2
D	5	3	6	5	m2
E	1	2	1	2	m1
F	4	2	4	3	m2
G	1	1	0	1	m1
H	2	1	1	0	m2



Exercise – Iteration 1

[STEP 4] The results obtained, members **cluster1 = {A, E, G}**, cluster2 = {B, C, D, F, H}. The SSE(Sum of Square Error) and MSE(Mean Square Error) values are:

$$SSE = \sum_{i=1}^k \sum_{p \in C_i} d(p, m_i)^2$$

$$\begin{aligned} SSE = & [(1-1)^2 + (3-1)^2] + [(1-1)^2 + (2-1)^2] + [(1-1)^2 + (1-1)^2] \\ & + [(3-2)^2 + (3-1)^2] + [(4-2)^2 + (3-1)^2] + [(5-2)^2 + (3-1)^2] \\ & + [(4-2)^2 + (2-1)^2] + [(2-2)^2 + (1-1)^2] \end{aligned}$$

$$SSE = 4 + 1 + 0 + 5 + 8 + 13 + 5 + 0 = 36$$

$$MSE = 36/8 = 4.5$$

Cluster1	Cluster 2
(1, 3)	(3, 3)
(1, 2)	(4, 3)
(1, 1)	(5, 3)
	(4, 2)
	(2, 1)

(3/3, 6/3)	(18/5, 12/5)
(1, 2)	(3.6, 2.4)

[STEP 5] The new **centroid** for

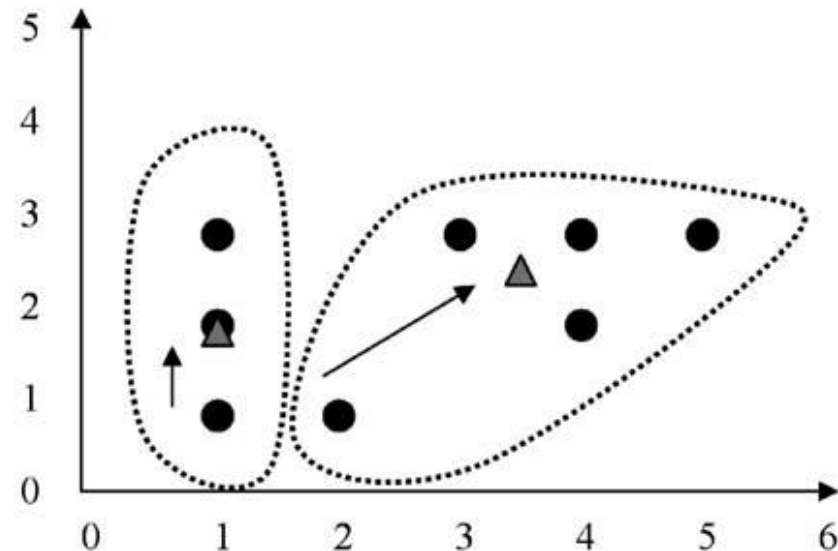
$$\text{cluster1} = [(1 + 1 + 1)/3, (3 + 2 + 1)/3] = (1, 2) \text{ and}$$

$$\text{cluster2} = [(3 + 4 + 5 + 4 + 2)/5, (3 + 3 + 3 + 2 + 1)/5] = (3.6, 2.4)$$

Exercise – Iteration 2

[STEP 3] Place each object to the closest cluster centroid based on the closest value difference (distance).

Instances	x	y	m1	m2	Cluster Membership
			(1,2)	(3.6,2.4)	
A	1	3	1	3.2	m1
B	3	3	3	1.2	m2
C	4	3	4	1	m2
D	5	3	5	2	m2
E	1	2	0	3	m1
F	4	2	3	0.8	m2
G	1	1	1	4	m1
H	2	1	2	3	m1



Exercise – Iteration 2

[STEP 4] The results obtained, members cluster1 = {A, E, G, H}, cluster2 = {B, C, D, F}. The new SSE and MSE values are:

$$SSE = \sum_{i=1}^k \sum_{p \in C_i} d(p, m_i)^2$$

$$\begin{aligned} SSE = & [(1-1)^2 + (3-2)^2] + [(1-1)^2 + (2-2)^2] + [(1-1)^2 + (1-2)^2] + [(2-1)^2 + (1-2)^2] \\ & + [(3-3.6)^2 + (3-2.4)^2] + [(4-3.6)^2 + (3-2.4)^2] + [(5-3.6)^2 + (3-2.4)^2] \\ & + [(4-3.6)^2 + (2-2.4)^2] \end{aligned}$$

$$SSE = 1 + 0 + 1 + 2 + 0.72 + 0.52 + 2.32 + 0.32 = 7.88$$

$$MSE = 7.88/8 = 0.985$$

[STEP 5] The new centroid for

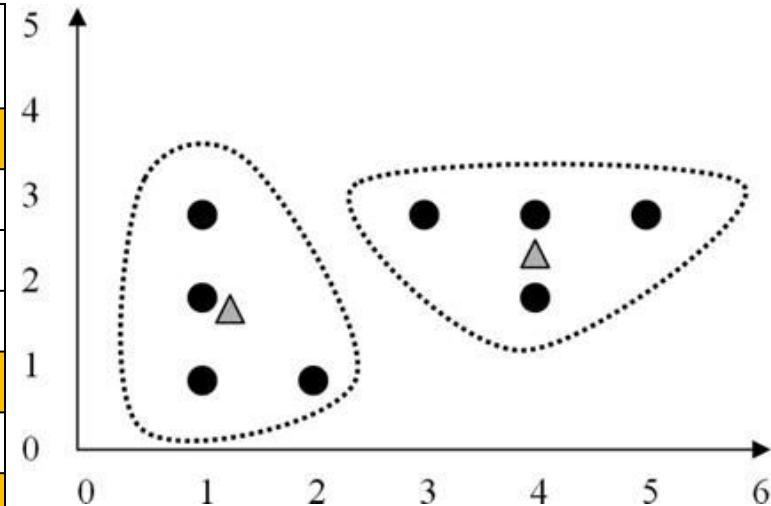
$$\text{cluster1} = [(1 + 1 + 1 + 2)/4, (3 + 2 + 1 + 1)/4] = (1.25, 1.75) \text{ and}$$

$$\text{cluster2} = [(3 + 4 + 5 + 4)/4, (3 + 3 + 3 + 2)/4] = (4, 2.75)$$

Exercise – Iteration 3

[STEP 3] Place each object to the closest cluster centroid based on the closest value difference (distance).

Instances	x	y	m1	m2	Cluster Membership
			(1.25,1.75)	(4, 2.75)	
A	1	3	1.5	3.25	m1
B	3	3	3	1.25	m2
C	4	3	4	0.25	m2
D	5	3	5	1.25	m2
E	1	2	0.5	3.75	m1
F	4	2	3	0.75	m2
G	1	1	1	4.75	m1
H	2	1	1.5	3.75	m1



Exercise – Iteration 3

[STEP 4] The results obtained, members cluster1 = {A, E, G, H}, cluster2 = {B, C, D, F}. The new SSE and MSE values are:

$$SSE = \sum_{i=1}^k \sum_{p \in C_i} d(p, m_i)^2$$

$$SSE = [(1-1.25)^2 + (3-1.75)^2] + [(1-1.25)^2 + (2-1.75)^2] + [(1-1.25)^2 + (1-1.75)^2] + [(2-1.25)^2 + (1-1.75)^2] + [(3-4)^2 + (3-2.75)^2] + [(4-4)^2 + (3-2.75)^2] + [(5-4)^2 + (3-2.75)^2] + [(4-4)^2 + (2-2.75)^2]$$

$$SSE = 1.625 + 0.125 + 0.625 + 0.125 + 1.0625 + 0.0625 + 1.0625 + 0.5625 = 7.88$$

$$MSE = 6.25/8 = 0.78125$$

[STEP 5] The new centroid for

$$\text{cluster1} = [(1 + 1 + 1 + 2)/4, (3 + 2 + 1 + 1)/4] = (1.25, 1.75) \text{ and}$$

$$\text{cluster2} = [(3 + 4 + 5 + 4)/4, (3 + 3 + 3 + 2)/4] = (4, 2.75)$$

Final Result

- ❖ Can be seen in the table. **No change of members again on each cluster**
 - The final results are:
 - cluster1 = {A, E, G, H}, and
 - cluster2 = {B, C, D, F}
 - With SSE value = 6.25 and MSE value = 0.78125
 - the number of iterations 3

Comments on the *K-Means* Method

❖ Strength:

- *Efficient*: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, $k, t \ll n$.
- Comparing: PAM: $O(k(n-k)^2)$, CLARA: $O(ks^2 + k(n-k))$

❖ Comment: Often terminates at a *local optimal*

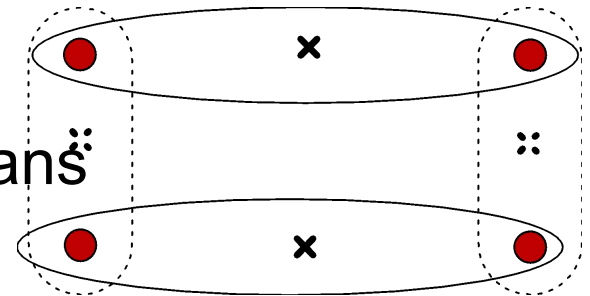
❖ Weakness

- Applicable only to **objects in a continuous n-dimensional** space
 - Using the k-modes method for categorical data
 - In comparison, k-medoids can be applied to a wide range of data
- **Need to specify k** , the *number* of clusters, in advance (there are ways to automatically determine the best k (see Hastie et al., 2009))
- **Sensitive to noisy data** and *outliers*
- Not suitable to discover clusters with *non-convex shapes*

Variations of the *K-Means* Method

❖ Most of the variants of the *k-means* which differ in

- Selection of the **initial *k* means**
- Dissimilarity calculations
- Strategies to calculate cluster means



❖ Handling categorical data: *k-modes*

- Replacing means of clusters with modes
- Using new dissimilarity measures to deal with categorical objects
- Using a frequency-based method to update modes of clusters
- A mixture of categorical and numerical data: *k-prototype* method

What Is the Problem of the K-Means Method?

- ❖ The k-means algorithm is **sensitive to outliers!**
 - Since an object with an extremely large value may substantially distort the distribution of the data
- ❖ K-Medoids:
 - Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster

